

Signal Processing

Exercise 8 – z-transform

SoSe 2026

I. Poles and zeros

- Zeros $z_{0,k}$: Numerator of transfer function $H(z)$ becomes zero
- Poles $z_{\infty,k}$: Denominator of transfer function $H(z)$ becomes zero
- Causality of a system
- Stability of a system
- Frequency response $H(e^{j\Omega})$ of a system

1. Causality

- Right-sided impulse response $h[n] \rightarrow$ depends only on current or past values of its input
- # poles $\geq$ # zeros; if # poles $<$ # zeros $\rightarrow$ system is non-causal

2. Stability

- Poles of a causal system are inside the unit circle $\rightarrow |z_{\infty, \max}| < 1$
- Region of convergence (ROC) includes unit circle
- ROC of a causal system is the area outside a circle with radius $r = |z_{\infty, k, \max}|$

3. Frequency response $H(e^{j\Omega})$ of a system

- From z -domain to frequency domain $\rightarrow$ set $z = e^{j\Omega}$ with $\Omega = 2\pi f \cdot T_s = 2\pi f / f_s$

Definition of the two-sided z -transform:

$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=-\infty}^{+\infty} x[n] \cdot z^{-n}$$

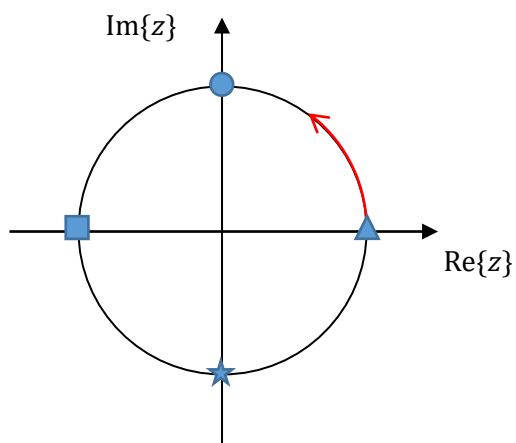
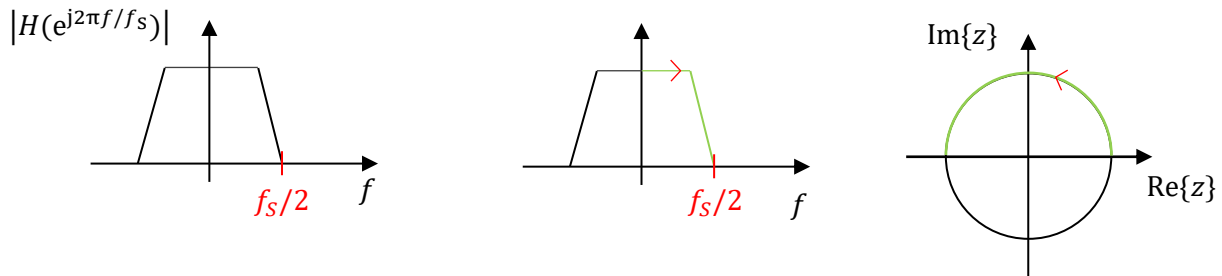
Definition of the DTFT :

- $X(e^{j\Omega})$ is periodic in 2π

$$X(e^{j\Omega}) = \mathcal{F}_*\{x[n]\} = \sum_{n=-\infty}^{+\infty} x[n] \cdot e^{-j\Omega n}$$

$\rightarrow$ DTFT is equal to the z -transform on the unit circle $\rightarrow$ frequency response $H(e^{j\Omega})$

II. Frequency response $H(e^{j\Omega})$; amplitude response $|H(e^{j\Omega})|$



- ▲ $z = 1 ; f = 0 ; f_A$; $\Omega = 0 ; 2\pi$
- $z = j ; f = f_A/4$; $\Omega = \pi/2$
- $z = -1 ; f = f_A/2$; $\Omega = \pi$
- ★ $z = -j ; f = 3/4 f_A$; $\Omega = 3/2\pi$

Question 1

The left hand side shows several pole-zero plots, corresponding amplitude responses $|H(e^{j\Omega})|$, which are not in the correct order, are shown on the right hand side (Fig. 1).

Assign the pole-zero plots to each corresponding magnitude response and specify the systems' filter characteristics.

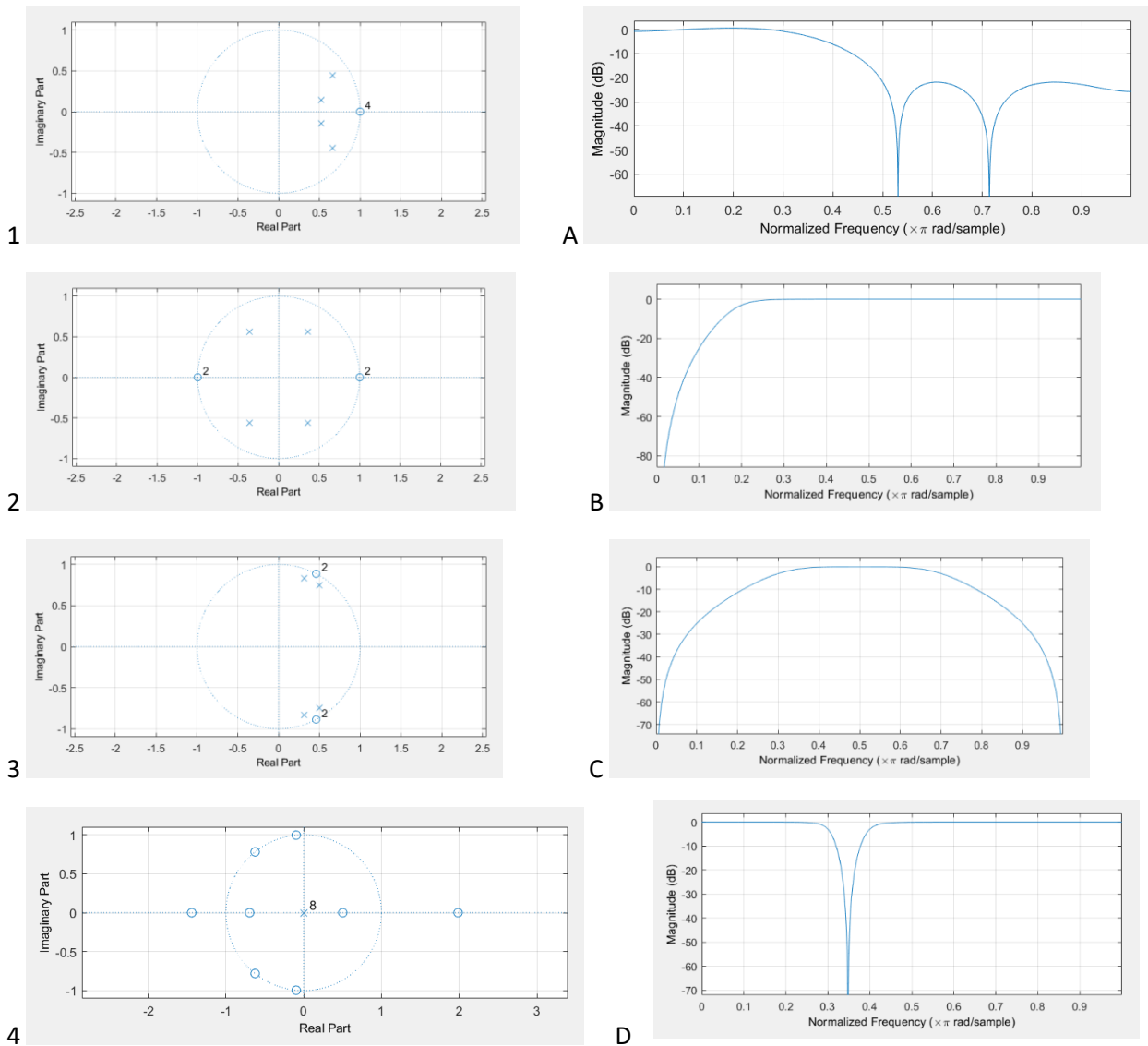


Figure 1: Pole-zero plot and corresponding, unsorted amplitude responses.

- 1 → Filter characteristic:
- 2 → Filter characteristic:
- 3 → Filter characteristic:
- 4 → Filter characteristic:

Question 2

A system is defined by the following difference equation:

$$y[n] = x[n] + x[n-1] - 0,1 \cdot y[n-1] + 0,3 \cdot y[n-2]$$

- Determine the transfer function $H(z)$.
- Sketch the corresponding pole-zero plot and draw the region of convergence (ROC) (Fig. 2).

Is the system stable, explain?

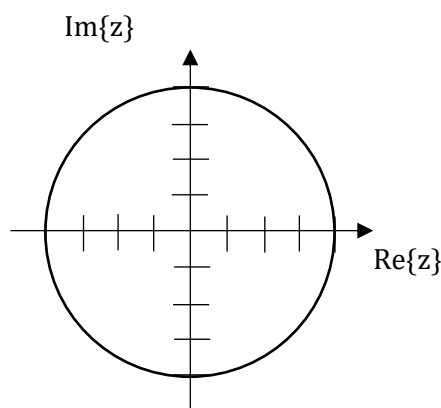


Figure 2: The system's pole-zero plot and its ROC.

- Is the system real / real valued?
- Determine the system's amplitude response $|H(e^{j\Omega})|$ and sketch it in the interval $\Omega = 0, \dots, 4\pi$ (Fig. 3). Define its filter characteristic.

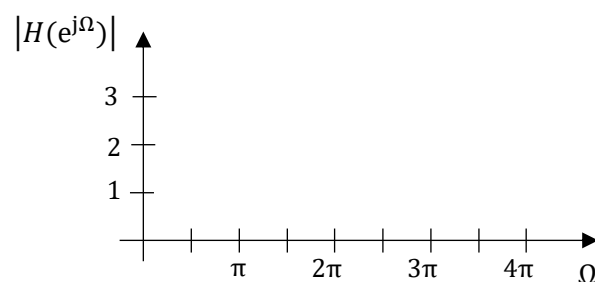


Figure 3: Amplitude response inside the interval $\Omega = 0, \dots, 4\pi$.

Question 3

A causal LTI system has a certain impulse response $h[n]$. Its transfer function $H(z)$ reads as

$$H(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) \cdot \left(1 + \frac{1}{4}z^{-1}\right)}$$

- Please sketch the poles and zeros (Fig. 4).
- Draw the system's ROC (Fig. 4). Is the system stable, explain!

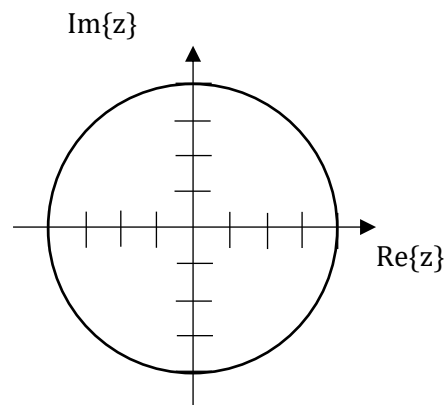


Figure 4: The system's pole-zero plot.

- Determine the impulse response $h[n]$.

Question 4

The impulse response $h[n]$ of a causal LTI system is defined

$$h[n] = \begin{cases} a^n, & \text{for } 0 \leq n \leq 7 \\ 0, & \text{otherwise} \end{cases}$$

- Draw the signal flow graph of the system in direct form (non-recursive implementation).
- Show that the corresponding transfer function can be expressed as

$$H(z) = \frac{1 - a^8 z^{-8}}{1 - a z^{-1}}, \quad |z| > |a|$$

Question 5

Consider a time-discrete LTI system that is described by the following difference equation:

$$y[n] = x[n] - 2x[n-1] + \frac{5}{2}y[n-1] - y[n-2]$$

- Determine the transfer function $H(z)$ in the frequency domain.
- Determine the impulse response $h[n]$.
- Is the system stable? Please give a short explanation.

Question 6

A time-discrete, causal system is defined by the following transfer function $H(z)$

$$H(z) = \frac{z+1}{z^2+5z+6}$$

- Determine the difference equation $y[n]$ of the system.
- Draw the system's block diagram with respect to a minimum order of delay elements.
- Please sketch the poles and zeros (Fig. 5). Is the system stable, explain!

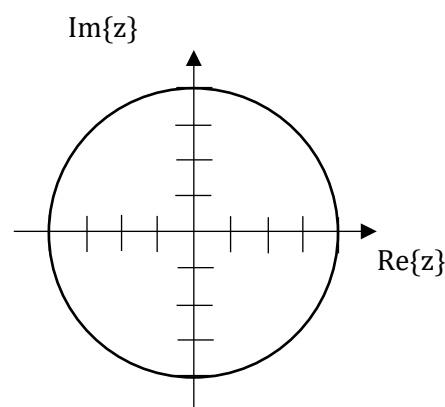


Figure 5: The system's pole-zero plot.

- Calculate the first five values of the system response $y[n]$ for $x[n] = \delta[n]$.
- Calculate the first five values of the system response $y[n]$ for $x[n] = 2\sigma[n]$.

Question 7

Consider the following block diagrams (Fig. 6).
Show that both diagrams represent the same discrete system.

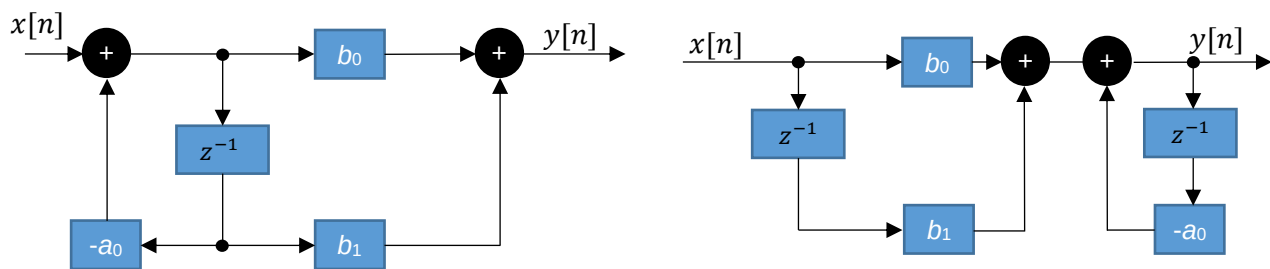


Figure 6: Block diagrams of the same discrete system.